

Architecture Laws for Machine Agency: A Synthesis of the Probe-Calibration Lineage

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Abstract

This note distills the program's practical architecture lessons from the first-order self, probe-calibration, world-responds, planning, and long-horizon papers. The empirical arc does not show consciousness, full agency, or general intelligence. It shows something narrower and more useful: small architectural changes repeatedly determine whether an agent's behavior is tied to the intended internal structure or to a proxy.

The recurring pattern is:

1. A behavior-level success appears.
2. A representation or control diagnostic shows the success is proxy-mediated.
3. A simple architecture change restores the intended causal structure.
4. A harder setting reveals the next proxy.

The resulting contribution is a ledger of architecture laws for machine agency: preserve vector-valued concern, use identifying interventions, recompute memory against the current model, normalize uncertainty before allocating inquiry, re-open inquiry after apparent convergence, cool at the decision layer rather than deleting the signal, split or gate heads when component identifiability stalls, and bind memory to future action surfaces.

1. Claim Boundary

These are **minimal computational precursors**. The experiments are mostly small homeostatic bandits, plus newer long-horizon language-agent diagnostics. The right claim is not "we made conscious agents." The right claim is that the program found a sequence of cheap, testable architecture constraints that move a system from behavior-only competence toward maintained self/world, memory/action, and concern/action coupling.

The recent non-peer-reviewed preprint by Lyons, Pio-Lopez, and Levin (2026) uses the phrase **virtual governor** for a distributed signal architecture that turns global constraint violations into local incentives. That phrase is useful framing here, but not evidence. In our terms, an architecture law is a tested constraint on the signal architecture: it asks whether the local action pressure really tracks the intended global concern.

2. The Ledger

Figure 2B. Architecture laws forced by the Metric Stack

The payoff is not one clever module; it is a reusable failure → law → payoff ledger.

Failure mode	Architecture law	Why it matters
Behavior-only success	Test internal load-bearing structure separately	Return and task success do not certify the intended representation.
Scalar priority collapse	Keep concern vector-valued until decision time	Priority shifts can reweight E and D without retraining the encoder.
Gauge-symmetric attribution	Use anchored interventions, not passive factorization	Null-anchor supervision breaks self/world false credit.
Stale or noisy uncertainty	Estimate current-model error on recent evidence	Probe calibration flips from anti-aligned to aligned.
Cross-dimension scale mismatch	Normalize uncertainty per concern dimension	Vector attribution becomes stable across seeds.
Anxiety or false calm	Cool at the decision layer, not the signal layer	The agent re-engages after shifts, then quiets after recovery.
Shared-head role ceiling	Split or gate heads when identifiability stalls	The next move is architectural or interventional, not more probing.
Long-horizon memory drift	Bind memory to commitment and tool-action surfaces	A later diagnostic turns future relevance into state/action sensitivity.

Failure mode	Architecture law	Agency payoff
Behavior-only success	Test internal load-bearing structure separately	Return is no longer mistaken for representation.
Scalar priority collapse	Keep concern vector-valued until decision time	Priorities can reweight E and D without retraining.
Gauge-symmetric attribution	Use anchored interventions, not passive factorization	Self/world false credit becomes measurable and correctable.
Stale or noisy uncertainty	Estimate current-model error on recent raw evidence	Probe calibration flips from anti-aligned to aligned.
Cross-dimension scale mismatch	Normalize uncertainty per concern dimension	Vector attribution becomes stable across seeds.
Learned quiet after shifts	Add a re-engagement floor or surprise detector	The agent can restart inquiry after the world changes.
Anxiety or false calm	Cool at the decision layer, not the signal layer	The agent re-engages after shifts and quiets after recovery.
Shared-head role ceiling	Split or gate heads when identifiability stalls	The next move becomes architectural or interventional.
Long-horizon memory drift	Bind memory to commitment and tool-action surfaces	Memory becomes control-relevant, not merely present.

3. Why These Laws Matter

3.1 Memory

The current-error calibration paper gives the simplest memory rule: do not store only stale residual tokens. Store enough raw evidence to replay it through the current model. A long-horizon agent's memory is useful when it can be revalued by the policy and world model that will act now.

3.2 Planning

Planning from concern closes a loop: learned concern-shaped representation can drive a predictive policy without external reward supervision. The architecture lesson is that a planner should consume a representation that preserves the dimensions of mattering until decision time. Premature scalarization makes planning brittle under shifted priorities.

3.3 Action

The null-intervention and world-responds papers show that self/world attribution requires actions whose purpose is identification, not reward. In nonstationary worlds, the action policy also needs a way to re-open inquiry after the model becomes quiet. Learned silence is not evidence of world stability.

3.4 Reafferece And Selfhood

The first-order self result is a negative law: input factorization does not identify causal source. Reafferece needs a gauge-breaking signal. In this program, the gauge breaker is an active null-anchor intervention plus calibrated probe selection. This is a selfhood ingredient, not a consciousness certificate.

3.5 Long Trajectories

The long-horizon bottleneck work adds the temporal complement: a hidden clue matters only if it reaches a later commitment surface, such as a tool call, schema field, or irreversible action. Memory without future action coupling is just context. Memory with commitment coupling becomes agentic state.

4. Simple Changes That Moved Outcomes

The strongest "small changes, large effects" observed so far are:

1. **Current replay:** keep a compact raw buffer and recompute residuals against the present model. This changed V_{probe} from anti-calibrated to positively calibrated.
2. **Per-dimension normalization:** scale-normalize probe targets or thresholds. This removed the vector seed-collapse failure.
3. **Active null anchoring:** add an identifying intervention instead of relying on passive self/world factorization. This broke the attribution gauge.
4. **Re-engagement floor / surprise trigger:** add a path back into inquiry after convergence. This fixes self-confirming silence.
5. **Decision-layer cooling:** damp the action tendency after recent probing without erasing the surprise signal. This avoids both anxiety and false calm.
6. **Commitment-surface memory:** test whether early information causally reaches a later action surface. This relocates long-horizon agency from context length to state/action coupling.

5. Major Contribution Opportunity

The major contribution is not a single module. It is a transferable method:

For every proposed agent capability, find the proxy that lets behavior pass without the intended structure, then add the smallest architecture change that makes the structure load-bearing.

This is where the work should push hardest next. The homeostatic arc produced the laws. The next contribution is to test whether they transfer to:

- long-horizon tool agents,
- structure-compatible OOD generalization,
- multi-agent or decentralized coordination,
- learned intervention discovery rather than hand-coded null actions,
- richer memory systems whose retrieval is coupled to future commitments.

6. Critical Caveats

- The homeostatic experiments use simulator-defined viability variables and a privileged null action.
- Most results use three seeds.
- The virtual-governor preprint is useful framing but not peer-reviewed evidence.
- The laws are empirically grounded within this repo's tasks; transfer to LLM agents, robotics, markets, or biological systems remains a hypothesis.
- Consciousness language must remain bounded. These laws concern control, attribution, memory, and inquiry surfaces, not subjective experience.

References

- Lyons, B., Pio-Lopez, L., & Levin, M. (2026). *Alignment is to a virtual governor: A theory of coordination in diverse intelligence*. Preprints.org. doi:10.20944/preprints202607.0220.v1. Not peer-reviewed.
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